# BANKING LOAN RISK ANALYSIS

# REST API — Technical Reference

Version 1.0.0 | OpenAPI 3.1.0

Generated from the supplied FastAPI OpenAPI specification.

Scope: authentication, customer records, credit activity, rule-based risk results, portfolio analytics, exports, and service health.

Base URL (local development): http://127.0.0.1:8000

# 1. API Overview

REST API for customer credit profiles, repayment behaviour, risk assessments, and portfolio analytics.

Interactive API explorer: http://127.0.0.1:8000/docs

Machine-readable specification: http://127.0.0.1:8000/openapi.json

Protected endpoints use an HTTP Bearer token. POST /auth/login accepts JSON credentials and returns an access_token. Supply Authorization: Bearer <token> on protected requests.

Roles used by the application: ADMIN, ANALYST, VIEWER. Customer creation, update, and deletion are restricted to ADMIN; CSV export is available to ADMIN and ANALYST. Other protected reads require authentication. These permissions come from the project implementation context; the OpenAPI security scheme alone does not encode role-level authorization.

# 2. Endpoint Index

| Method | Path | Purpose | Auth |
|---|---|---|---|
| GET | /health | Health Check | Public |
| GET | /customers | List Customers | Bearer |
| POST | /customers | Create New Customer | Bearer |
| GET | /customers/{customer_id} | Customer Detail | Bearer |
| PUT | /customers/{customer_id} | Update Existing Customer | Bearer |
| DELETE | /customers/{customer_id} | Delete Existing Customer | Bearer |
| GET | /customers/{customer_id}/activity | Customer Activity | Bearer |
| GET | /customers/{customer_id}/risk | Customer Risk | Bearer |
| GET | /analytics/summary | Portfolio Summary | Bearer |
| GET | /analytics/risk-distribution | Risk Distribution | Bearer |
| GET | /analytics/monthly-trends | Monthly Trends | Bearer |
| GET | /analytics/segments | Segment Analytics | Bearer |
| GET | /export/customers | Export Customers | Bearer |
| POST | /auth/login | Login | Public |
| GET | /auth/me | Get Authenticated User | Bearer |
| GET | / | Root | Public |

# 3. Endpoint Reference

## GET /health

Health Check. Check API and database availability.

Authentication: Not required by OpenAPI specification

Declared responses: 200: Successful Response — Unspecified JSON schema

## GET /customers

List Customers.

Authentication: Bearer JWT required

| Parameter | Location | Type / allowed values | Required |
|---|---|---|---|
| page | query | integer; default: 1 | No |
| page_size | query | integer; default: 20 | No |
| risk_band | query | string/null; values: LOW, MEDIUM, HIGH | No |
| has_payment_delay | query | boolean/null | No |
| min_credit_limit | query | number/null | No |
| max_credit_limit | query | number/null | No |
| sort_by | query | string; values: customer_id, credit_limit, latest_utilization_ratio, delay_month_count, risk_score; default: customer_id | No |
| sort_order | query | string; values: asc, desc; default: asc | No |

Declared responses: 200: Successful Response — CustomerListResponse; 422: Validation Error — HTTPValidationError

## POST /customers

Create New Customer.

Authentication: Bearer JWT required

JSON request body: CustomerCreateRequest (see Section 4).

Declared responses: 201: Successful Response — CustomerMutationResponse; 422: Validation Error — HTTPValidationError

## GET /customers/{customer_id}

Customer Detail.

Authentication: Bearer JWT required

| Parameter | Location | Type / allowed values | Required |
|---|---|---|---|
| customer_id | path | integer | Yes |

Declared responses: 200: Successful Response — CustomerDetailResponse; 422: Validation Error — HTTPValidationError

## PUT /customers/{customer_id}

Update Existing Customer.

Authentication: Bearer JWT required

| Parameter | Location | Type / allowed values | Required |
|---|---|---|---|
| customer_id | path | integer | Yes |

JSON request body: CustomerUpdateRequest (see Section 4).

Declared responses: 200: Successful Response — CustomerMutationResponse; 422: Validation Error — HTTPValidationError

## DELETE /customers/{customer_id}

Delete Existing Customer.

Authentication: Bearer JWT required

| Parameter | Location | Type / allowed values | Required |
|---|---|---|---|
| customer_id | path | integer | Yes |

Declared responses: 200: Successful Response — CustomerDeleteResponse; 422: Validation Error — HTTPValidationError

## GET /customers/{customer_id}/activity

Customer Activity.

Authentication: Bearer JWT required

| Parameter | Location | Type / allowed values | Required |
|---|---|---|---|
| customer_id | path | integer | Yes |

Declared responses: 200: Successful Response — CustomerActivityResponse; 422: Validation Error — HTTPValidationError

## GET /customers/{customer_id}/risk

Customer Risk.

Authentication: Bearer JWT required

| Parameter | Location | Type / allowed values | Required |
|---|---|---|---|
| customer_id | path | integer | Yes |

Declared responses: 200: Successful Response — CustomerRiskResponse; 422: Validation Error — HTTPValidationError

## GET /analytics/summary

Portfolio Summary.

Authentication: Bearer JWT required

Declared responses: 200: Successful Response — PortfolioSummaryResponse

## GET /analytics/risk-distribution

Risk Distribution.

Authentication: Bearer JWT required

Declared responses: 200: Successful Response — RiskDistributionItem[]

## GET /analytics/monthly-trends

Monthly Trends.

Authentication: Bearer JWT required

Declared responses: 200: Successful Response — MonthlyTrendItem[]

## GET /analytics/segments

Segment Analytics.

Authentication: Bearer JWT required

| Parameter | Location | Type / allowed values | Required |
|---|---|---|---|
| segment | query | string; values: gender, education, marital_status, age | Yes |

Declared responses: 200: Successful Response — SegmentAnalyticsItem[]; 422: Validation Error — HTTPValidationError

## GET /export/customers

Export Customers.

Authentication: Bearer JWT required

| Parameter | Location | Type / allowed values | Required |
|---|---|---|---|
| risk_band | query | string/null; values: LOW, MEDIUM, HIGH | No |
| has_payment_delay | query | boolean/null | No |
| min_credit_limit | query | number/null | No |
| max_credit_limit | query | number/null | No |

| Parameter | Location | Type / allowed values | Required |
|---|---|---|---|
| sort_by | query | string; values: customer_id, credit_limit, latest_utilization_ratio, delay_month_count, risk_score; default: customer_id | No |
| sort_order | query | string; values: asc, desc; default: asc | No |

Declared responses: 200: Successful Response — Unspecified JSON schema; 422: Validation Error — HTTPValidationError

### POST /auth/login

Login.

Authentication: Not required by OpenAPI specification

JSON request body: LoginRequest (see Section 4).

Declared responses: 200: Successful Response — LoginResponse; 422: Validation Error — HTTPValidationError

### GET /auth/me

Get Authenticated User. Return the currently authenticated user's profile.

Authentication: Bearer JWT required

Declared responses: 200: Successful Response — CurrentUserResponse

### GET /

Root.

Authentication: Not required by OpenAPI specification

Declared responses: 200: Successful Response — Unspecified JSON schema

## 4. Request and Response Models

The following fields are taken directly from the OpenAPI schema. Required fields are marked with *. Nullable values are indicated where declared.

### CurrentUserResponse

| Field | Type / constraints |
|---|---|
| user_id* | integer |
| username* | string |
| email* | string |
| role* | string [ADMIN, ANALYST, VIEWER] |

### CustomerActivityResponse

| Field | Type / constraints |
|---|---|
| customer_id* | integer |
| activity* | array of MonthlyActivityItem |

### CustomerCreateRequest

| Field | Type / constraints |
|---|---|
| customer_id* | integer; min 1.0 |

| Field | Type / constraints |
|---|---|
| gender_code* | integer [1, 2] |
| education_code* | integer [0, 1, 2, 3, 4, 5, 6] |
| marital_status_code* | integer [0, 1, 2, 3] |
| age* | integer |
| credit_limit* | number |
| default_next_month* | integer [0, 1] |
| monthly_activity* | array of MonthlyActivityCreate; items 6-6 |

## CustomerCreditProfile

| Field | Type / constraints |
|---|---|
| credit_limit* | number |
| delay_month_count* | integer |
| max_delay_status* | integer |
| has_payment_delay* | boolean |
| total_bill_amount_6m* | number |
| average_bill_amount_6m* | number |
| total_payment_amount_6m* | number |
| average_payment_amount_6m* | number |
| latest_utilization_ratio* | number |
| default_next_month* | integer |

## CustomerDeleteResponse

| Field | Type / constraints |
|---|---|
| message* | string |
| customer_id* | integer |

## CustomerDemographics

| Field | Type / constraints |
|---|---|
| gender* | string |
| education* | string |
| marital_status* | string |
| age* | integer |

## CustomerDetailResponse

| Field | Type / constraints |
|---|---|
| customer_id* | integer |
| demographics* | CustomerDemographics |
| credit_profile* | CustomerCreditProfile |
| risk* | CustomerRiskAssessment |

## CustomerListItem

| Field | Type / constraints |
|---|---|
| customer_id* | integer |
| age* | integer |
| gender* | string |
| education* | string |
| marital_status* | string |
| credit_limit* | number |
| has_payment_delay* | boolean |
| delay_month_count* | integer |
| latest_utilization_ratio* | number |
| risk_score* | integer |
| risk_band* | string |

## CustomerListResponse

| Field | Type / constraints |
|---|---|
| page* | integer |
| page_size* | integer |
| total_records* | integer |
| total_pages* | integer |
| items* | array of CustomerListItem |

## CustomerMutationResponse

| Field | Type / constraints |
|---|---|
| message* | string |
| customer_id* | integer |
| risk_score* | integer |

| Field | Type / constraints |
| --- | --- |
| risk_band* | string |

## CustomerRiskAssessment

| Field | Type / constraints |
| --- | --- |
| payment_to_bill_ratio* | number / null |
| delay_frequency_score* | integer |
| delay_severity_score* | integer |
| utilization_score* | integer |
| payment_behavior_score* | integer |
| risk_score* | integer |
| risk_band* | string |
| rule_version* | string |

## CustomerRiskResponse

| Field | Type / constraints |
| --- | --- |
| customer_id* | integer |
| payment_to_bill_ratio* | number / null |
| delay_frequency_score* | integer |
| delay_severity_score* | integer |
| utilization_score* | integer |
| payment_behavior_score* | integer |
| risk_score* | integer |
| risk_band* | string |
| rule_version* | string |

## CustomerUpdateRequest

| Field | Type / constraints |
| --- | --- |
| gender_code* | integer [1, 2] |
| education_code* | integer [0, 1, 2, 3, 4, 5, 6] |
| marital_status_code* | integer [0, 1, 2, 3] |
| age* | integer |
| credit_limit* | number |
| default_next_month* | integer [0, 1] |

| Field | Type / constraints |
|---|---|
| monthly_activity* | array of MonthlyActivityCreate; items 6-6 |

## HTTPValidationError

| Field | Type / constraints |
|---|---|
| detail | array of ValidationError |

## LoginRequest

| Field | Type / constraints |
|---|---|
| username* | string |
| password* | string |

## LoginResponse

| Field | Type / constraints |
|---|---|
| access_token* | string |
| token_type | string |
| expires_in* | integer |
| user_id* | integer |
| username* | string |
| role* | string [ADMIN, ANALYST, VIEWER] |

## MonthlyActivityCreate

| Field | Type / constraints |
|---|---|
| activity_month* | string |
| repayment_status* | integer; min -2.0; max 8.0 |
| bill_amount* | number |
| payment_amount* | number; min 0.0 |

## MonthlyActivityItem

| Field | Type / constraints |
|---|---|
| activity_month* | string |
| repayment_status* | integer |
| bill_amount* | number |
| payment_amount* | number |

## MonthlyTrendItem

| Field | Type / constraints |
|---|---|
| activity_month* | string |
| customer_records* | integer |
| total_bill_amount* | number |
| average_bill_amount* | number |
| total_payment_amount* | number |
| average_payment_amount* | number |
| delayed_customers* | integer |
| delayed_customer_rate_pct* | number |

## PortfolioSummaryResponse

| Field | Type / constraints |
|---|---|
| total_customers* | integer |
| total_credit_exposure* | number |
| average_credit_limit* | number |
| historical_defaults* | integer |
| historical_default_rate_pct* | number |
| customers_with_payment_delay* | integer |
| customers_with_repeated_delay* | integer |
| average_latest_utilization_pct* | number |
| over_limit_customers* | integer |
| total_bill_amount_6m* | number |
| total_payment_amount_6m* | number |

## RiskDistributionItem

| Field | Type / constraints |
|---|---|
| risk_band* | string |
| total_customers* | integer |
| historical_defaults* | integer |
| historical_default_rate_pct* | number |
| average_risk_score* | number |

### SegmentAnalyticsItem

| Field | Type / constraints |
|---|---|
| segment* | string |
| total_customers* | integer |
| average_credit_limit* | number |
| historical_defaults* | integer |
| historical_default_rate_pct* | number |
| customers_with_payment_delay* | integer |
| average_latest_utilization_pct* | number |

### ValidationError

| Field | Type / constraints |
|---|---|
| loc* | array of items |
| msg* | string |
| type* | string |
| input | |
| ctx | object |

## 5. Authentication and Access

Login: POST /auth/login with username and password as JSON. The response includes access_token, token_type (bearer), expires_in, user_id, username, and role. GET /auth/me returns the authenticated user profile.

`Example login request: {"username":"example_admin","password":"<password>"}`

`Example authenticated header: Authorization: Bearer <access_token>`

Never commit passwords, JWT secrets, or real access tokens to documentation or source control.

## 6. Validation and Errors

The OpenAPI specification explicitly documents HTTP 422 validation errors for routes with validated inputs. HTTP 200 is used for most successful requests and HTTP 201 for customer creation.

Authentication failures, forbidden operations, missing customers, database errors, and server errors may occur at runtime, but not all are enumerated as response objects in the supplied OpenAPI file. Do not treat the documented responses as an exhaustive error contract.

## 7. Notes and Limitations

The API returns stored deterministic risk assessments rather than predictions from a machine-learning model. The customer export endpoint is designed to return CSV content; the OpenAPI specification does not describe its CSV response schema explicitly.

The OpenAPI file defines 13 unique paths and 17 HTTP operations. This document is a reference for the supplied API version; regenerate it after endpoint or schema changes.